

Final Project Assignment

Exercise 1 - Baseline Synthesis (2 points)

Using the original `synthesis.tcl` script provided in `WORK_Group20.zip`, synthesize the given RTL design using the STcmos65 SVT library.

Note: The files `tech/synopsys_dc.setup` and `tech/synopsys_pt.setup` are not provided and must be implemented.

Find the minimum clock period, among values that are multiples of 0.5 ns, such that the following constraints are met, with both metrics measured in PrimeTime:

- Slack ≥ 0
- Area < 18000

Note: the `synthesis.tcl` script cannot be modified.

For the selected solution, extend the `pt_analysis.tcl` script to report the following metrics:

- Clock period (ns)
- Area (um2)
- Slack (ns)
- Leakage power (mW)
- Dynamic power (mW)

All metrics must be measured in PrimeTime. Power values must be measured using VCD-based SWA back-annotation.

Exercise 2 - Power Optimization (2 points)

2.1 Clock Gating

Starting from the clock period found in Exercise 1, synthesize the design with clock gating enabled.

Apply clock gating with the following settings:

- Minimum bit-width: 8
- Maximum fanout: 32

For the selected solution, extend the `pt_analysis.tcl` script to report the following metrics:

- Area (um2)
- Slack (ns)
- Leakage power (mW)
- Dynamic power (mW)

All metrics must be measured in PrimeTime. Power values must be measured using VCD-based SWA back-annotation. In the report, discuss and motivate the impact of clock gating on PPA metrics.

2.2 Clock Gating and Multi-VT Synthesis

Starting from the clock period found in Exercise 1, synthesize the design using both:

- Clock gating, with the same settings used in Exercise 2.1
- Multi-VT synthesis (with LVT, SVT, and HVT)

Note: use same input driving cells, operating conditions and wire load model as in Exercise 1.

For the obtained solution, extend the `pt_analysis.tcl` script to report the following metrics:

- Percentage of LVT cells
- Percentage of SVT cells
- Percentage of HVT cells
- Area (μm^2)
- Slack (ns)
- Leakage power (mW)
- Dynamic power (mW)

All metrics must be measured in PrimeTime. Power values must be measured using VCD-based SWA back-annotation. In the report, discuss and motivate the impact of Multi-VT synthesis on PPA metrics.

Suggestion: In a design with clock gating, use the following command to get the collection of all gates in the design:

```
set gates [get_cells -hierarchical -filter "is_hierarchical == false"]
```

Exercise 3 - Custom Post-Synthesis Report in PrimeTime (2 points)

Implement a custom PrimeTime procedure that performs pin-based critical-path analysis.

The procedure must take the following input arguments:

- `delay_type` (str): the timing analysis type. Accepted values are `min` and `max`.
- `num_paths` (int): the number of most critical paths to report.
- `pin_full_name` (str): the full name of a pin belonging to a combinational cell.

The procedure must print to the command-line interface a textual report listing the `num_paths` most critical paths that traverse `pin_full_name` under the specified `delay_type` analysis.

Each row of the report must contain:

`<startpoint>` `<endpoint>` `<slack>`

where:

- `<startpoint>` is the startpoint of the timing path.
- `<endpoint>` is the endpoint of the timing path.
- `<slack>` is the slack of the timing path.

Rows must be sorted by `<slack>` in ascending order.

Deliverables

Submit a zip file named `Group20.zip`.

The archive must contain four folders, one for each exercise, and one final report, following the structure below:

```
Group20/  
  ex1/
```

```

    siphash_core.sdc
    synopsys_dc.setup
    synopsys_pt.setup
    pt_analysis.tcl
ex2.1/
    siphash_core.sdc
    synopsys_dc.setup
    synopsys_pt.setup
    synthesis.tcl
    pt_analysis.tcl
ex2.2/
    siphash_core.sdc
    synopsys_dc.setup
    synopsys_pt.setup
    synthesis.tcl
    pt_analysis.tcl
ex3/
    custom_report.tcl
REPORT.pdf

```

Additional requirements:

- In `ex1/pt_analysis.tcl`, include the commands for SWA back-annotation and for reporting the requested metrics.
- In `ex2.1/synthesis.tcl`, include all clock-gating commands.
- In `ex2.1/pt_analysis.tcl`, include the commands for SWA back-annotation and for reporting the requested metrics.
- In `ex2.2/synthesis.tcl`, include all clock-gating and Multi-VT synthesis commands.
- In `ex2.2/pt_analysis.tcl`, include the commands for SWA back-annotation and for reporting the requested metrics.
- In `ex3/custom_report.tcl`, include the code of the requested TCL procedure.
- For `REPORT.pdf`, use the tex template available in `WORK_Group20.zip`.

Suggestions

- Review all lab slides and tutorials.
- Review the solutions to the lab exercises uploaded to the Material page.

Submission Instructions

- Each group must send an e-mail to `valentino.peluso@polito.it`, with `andrea.calimera@polito.it` in cc.
- Use the following subject: `SODS26 Group20`.
- Attach the file `Group20.zip` containing all required deliverables.

Deadline: July 7, 23:59

Late submissions, or submissions not compliant with the specifications above, will be automatically discarded.

